



23MEE384
INDUSTRIAL AUTOMATION - Laboratory

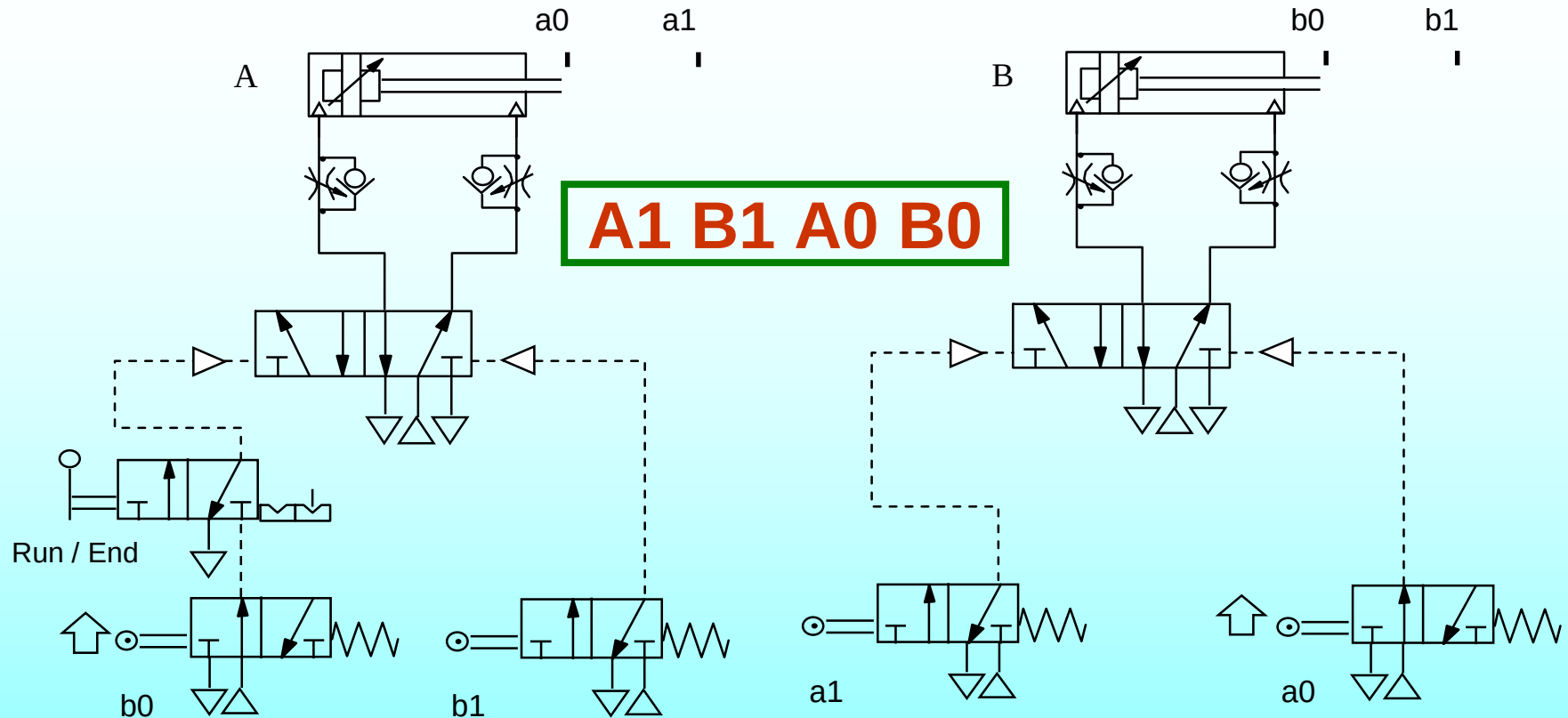
AMRITA SCHOOL OF ENGINEERING
DEPARTMENT OF MECHANICAL
ENGINEERING
COIMBATORE CAMPUS

HYDRAULIC / PNEUMATIC CIRCUIT DESIGN – CASCADE METHOD

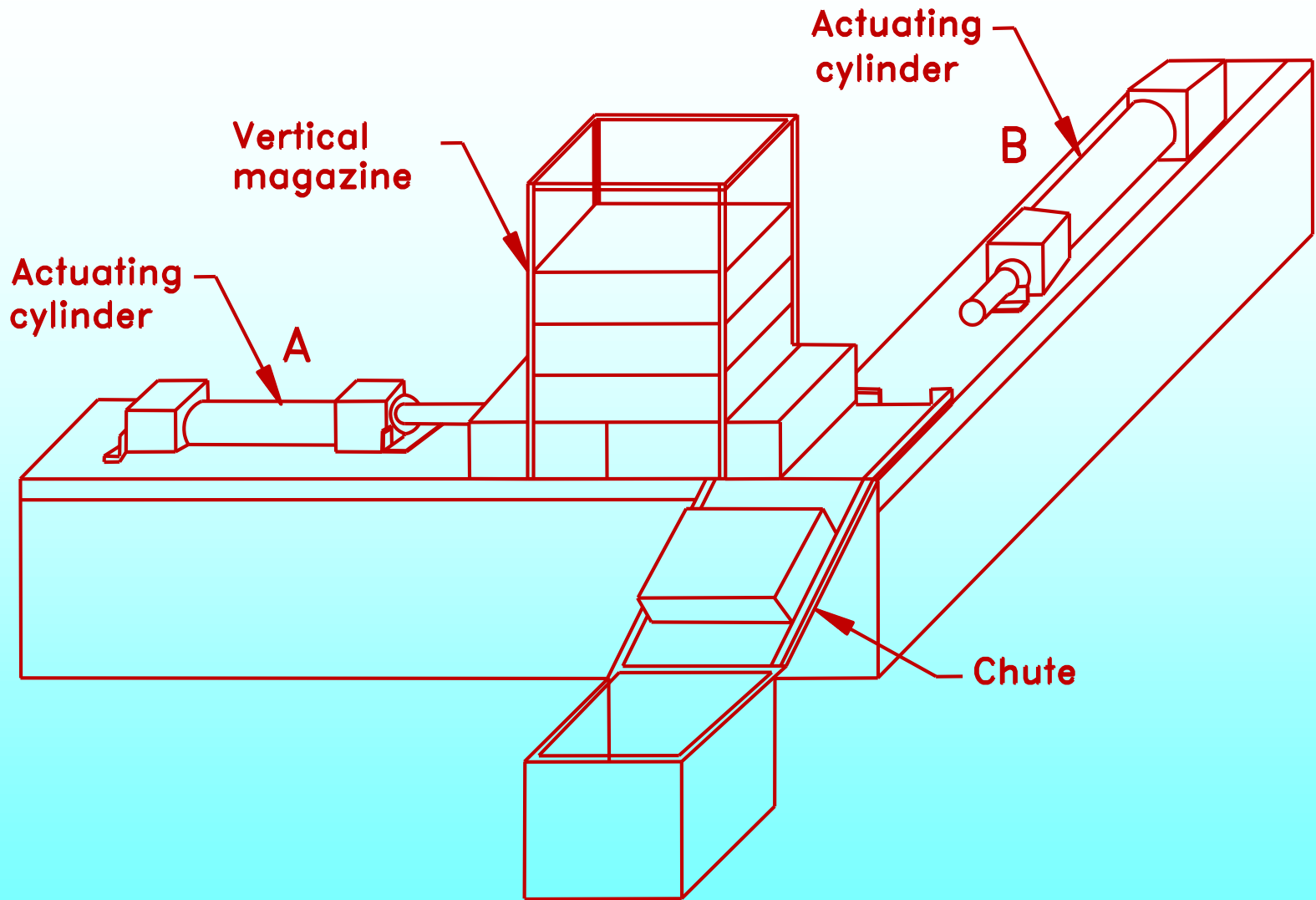
REPEAT PATTERN SEQUENCE

- A repeat pattern sequence is one where the order of the movements in the first half of the sequence is repeated in the second half
 - Each actuator may have one Out and In stroke only in the sequence
 - There may be any number of actuators in the sequence
- The signal starting the first movement must pass through the Run/End valve
 - Needs only the basic building blocks to solve
 - Examples of repeat pattern sequences:
 - A+ B+ C+ D+ A- B- C- D-
 - A- B+ C- A+ B- C+
 - C+ A+ B- C- A- B+

REPEAT PATTERN SEQUENCE



- The two cylinders A and B are to perform a simple repeat pattern sequence as follows: A+ B+ A- B-
- Apply the rule “The signal given by the completion of each movement will initiate the next movement”
- In this way the roller valves can be identified and labelled



Configuration of transfer station

PNEUMATIC TRANSFER STATION

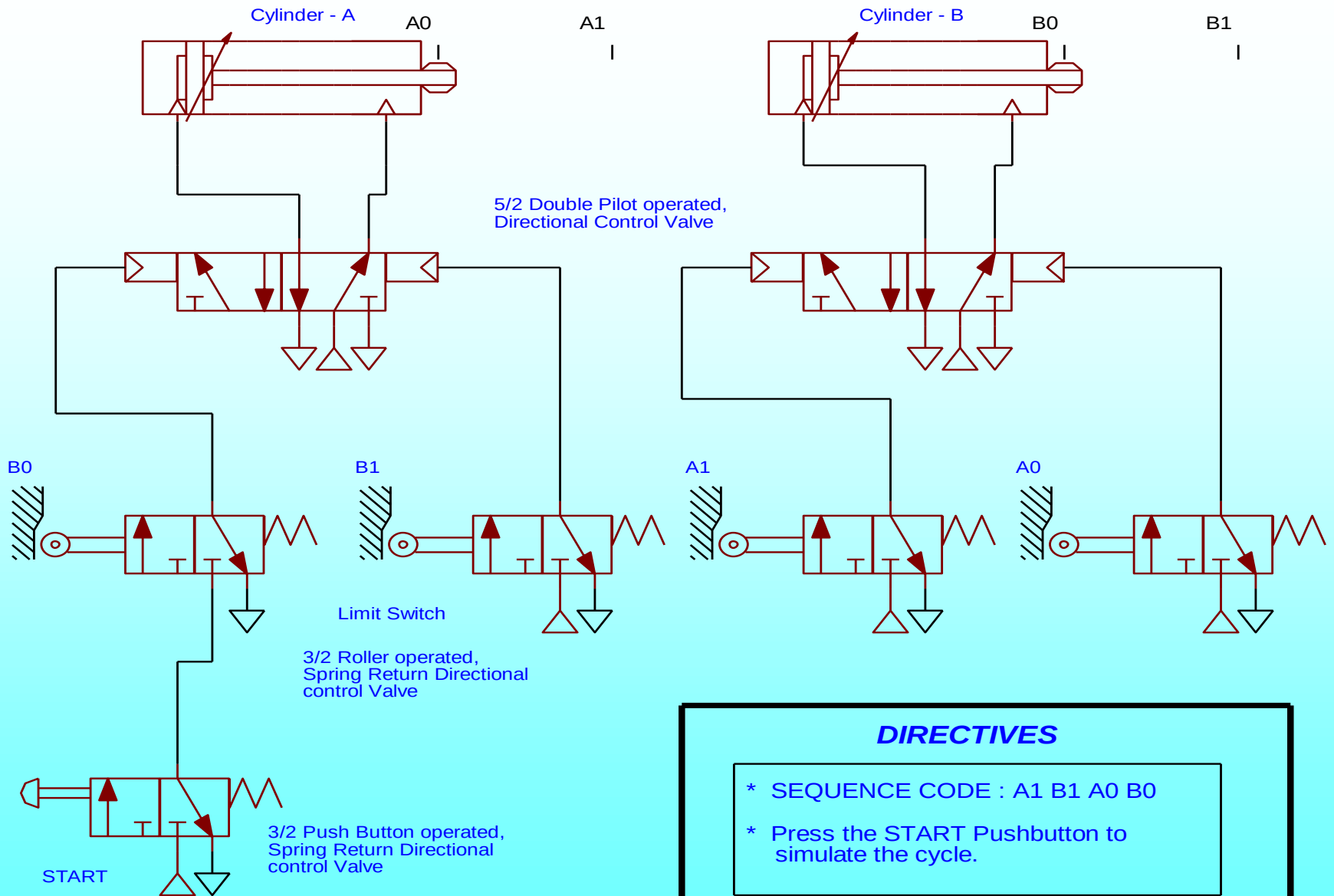
Problem:

Two cylinders are used to transfer parts from a magazine onto a chute. When a pushbutton is pressed, cylinder (A) extends, pushing the part from the magazine and positions it in preparation for transfer by cylinder (B) onto the outfeed chute. Once the part is transferred, the cylinder (A) retracts, followed by the cylinder (B).

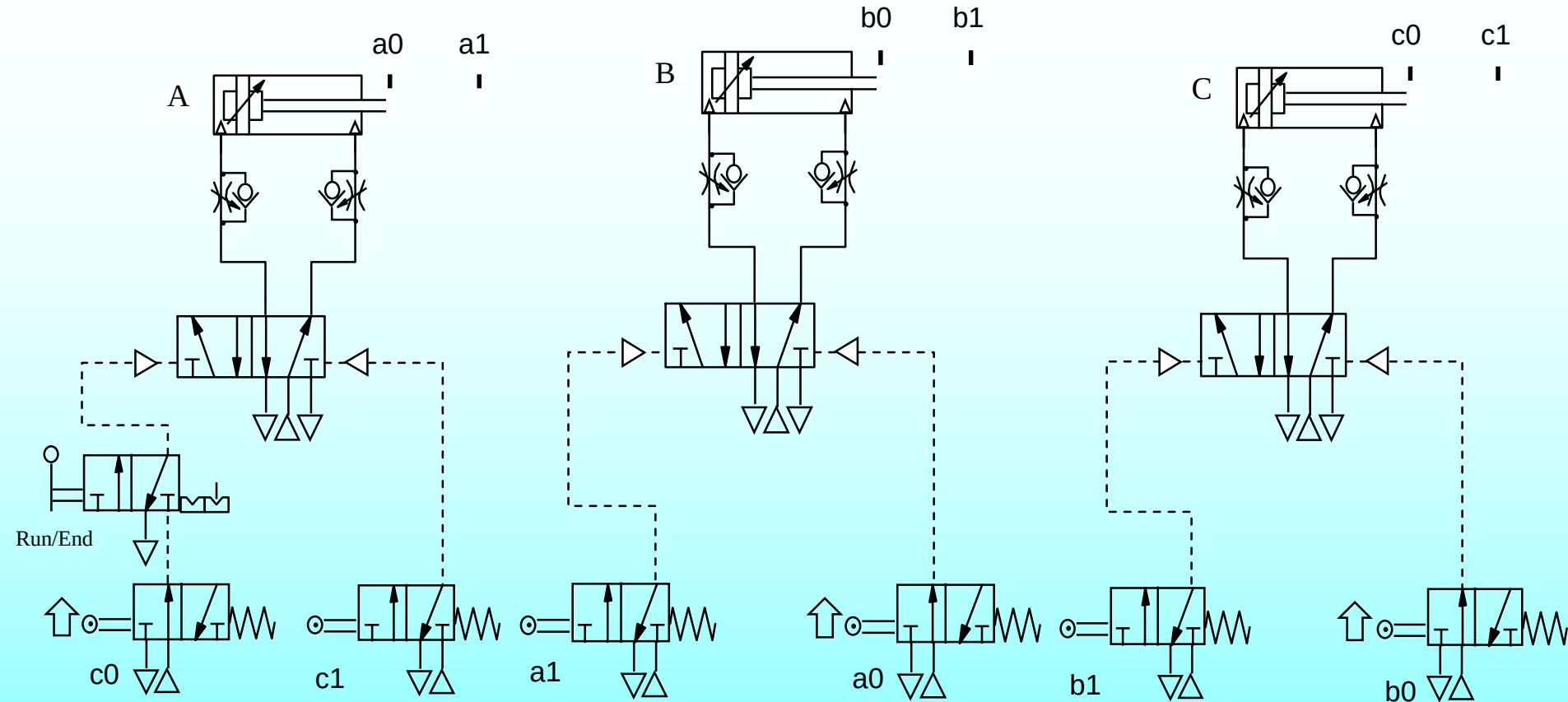
POSSIBLE SEQUENCE OF OPERATIONS ARE:

A1 A0 B1 B0 ; A1 B1 A0 B0 ;

A1 B1 B0 A0 ;



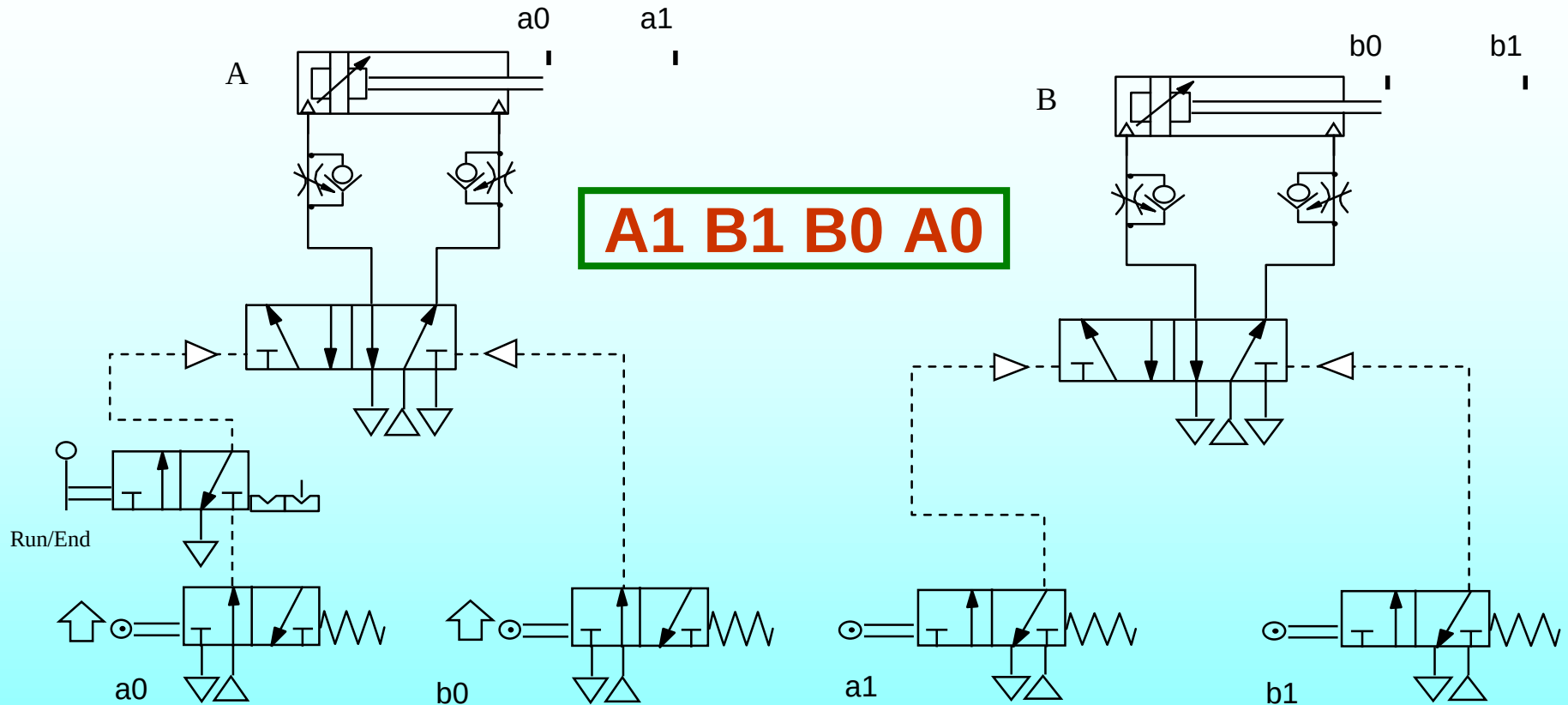
REPEAT PATTERN SEQUENCE



A1 B1 C1 A0 B0 C0

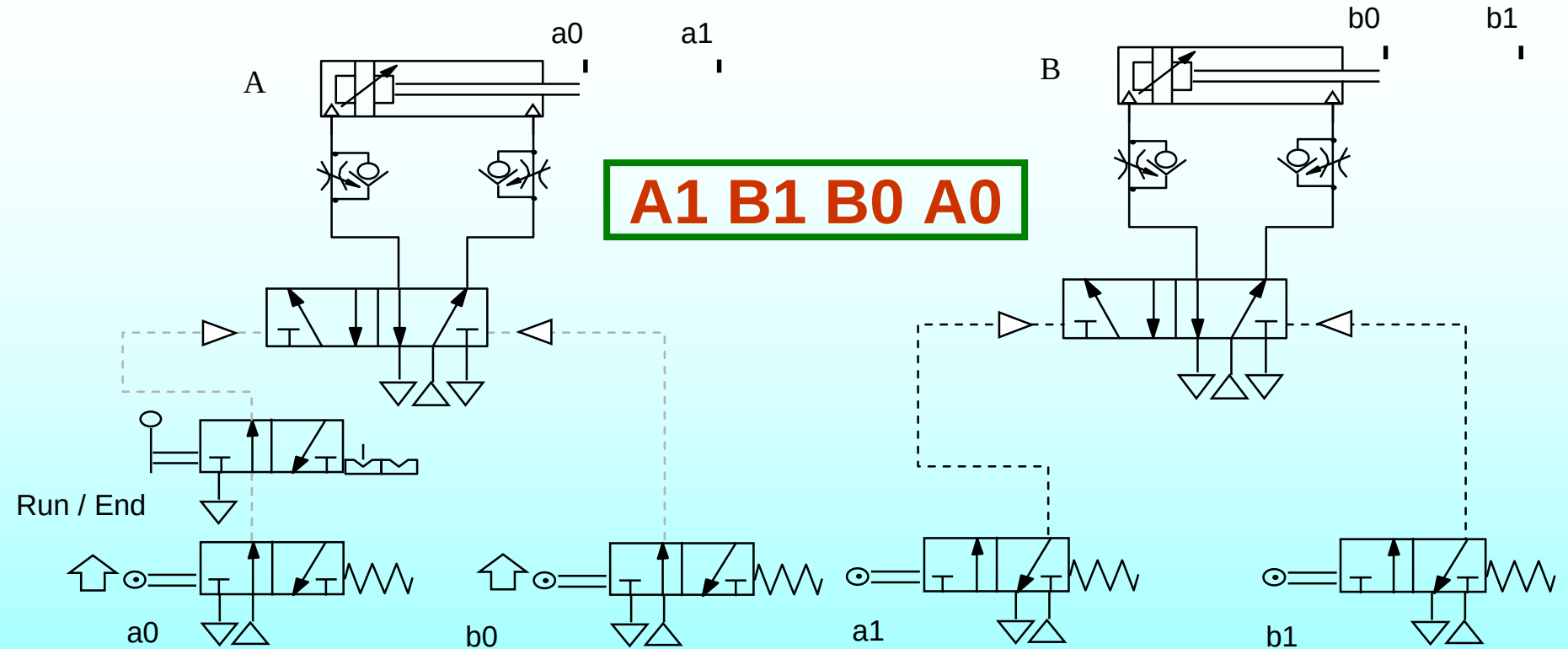
- For three cylinders A, B and C also to perform a simple repeat pattern sequence as follows: A+ B+ C+ A- B- C-
- Apply the rule “The signal given by the completion of each movement will initiate the next movement”

NON - REPEAT PATTERN SEQUENCE



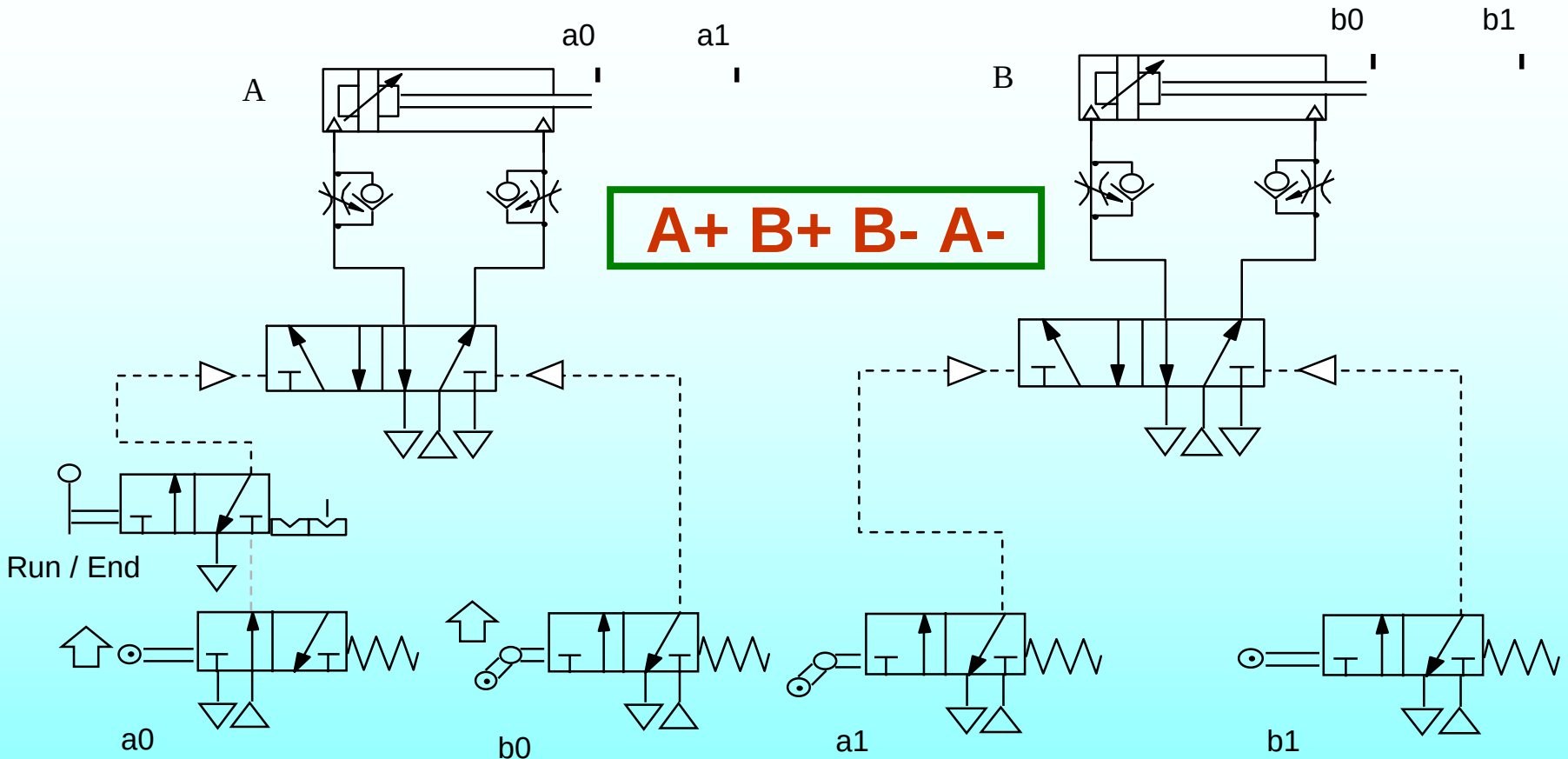
- If the rule applied to a repeat pattern sequence is applied to any other sequence there will be opposed signals on one or more of the 5/2 valves preventing operation
- This circuit demonstrates the problem
- The sequence is A+ B+ B- A-

OPPOSED SIGNALS



- When the valve is set to Run, cylinder A will not move because the 5/2 valve has an opposed signal, it is still being signalled to hold position by the feedback valve b0
- If A was able to move + a similar problem will occur for the 5/2 valve of B once it was +
- The sequence is A+ B+ B- A-

MECHANICAL SOLUTION



- The problem was caused by valves b0 and a1 being operated at the time the new opposing instruction is given
- If these two valves were “one way trip” types and over tripped at the last movement of stroke, only a pulse would be obtained instead of a continuous signal

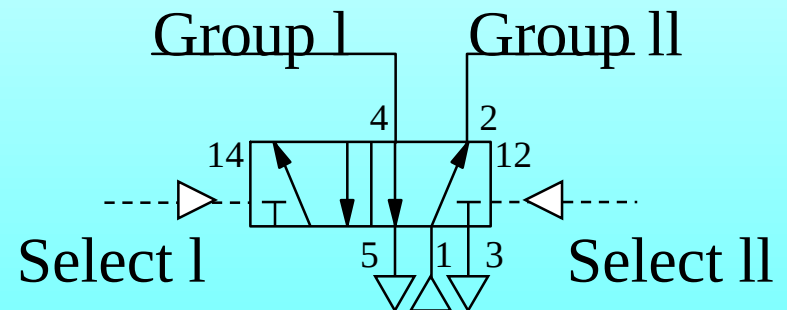
SEQUENCE SOLUTION METHODS

- The main solutions to solving sequences are:
 - Cascade (pneumatic & Hydraulic)
 - Shift register (pneumatic & Hydraulic)
 - Electro-pneumatic or Electro-hydraulic
 - PLC (Programmable logic controller)
- Cascade circuits provide a standard method of solving any sequence. It uses a minimum of additional logic hardware (one logic valve per group of sequential steps)
- Shift register circuits are similar to cascade but use one logic valve for every step
- Electro-pneumatic circuits use solenoid valves and electro-mechanical relays
- PLC. The standard solution for medium to complex sequential systems (except where electrical equipment cannot be used)

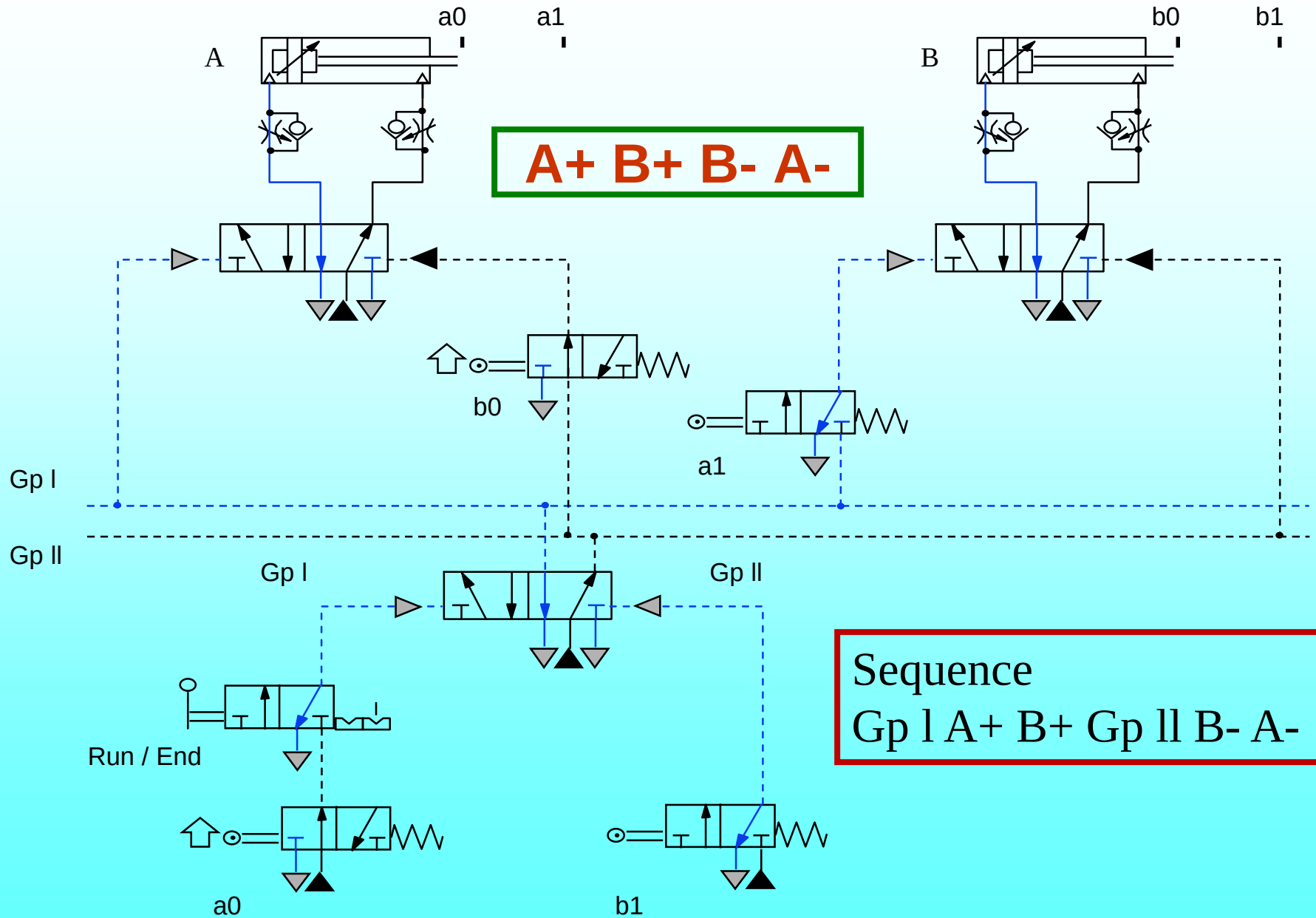
CASCADE TWO GROUP

- The A+ B+ B- A- circuit is solved by the two group cascade method
- The sequence is divided at the point where B immediately returns
- The two parts are allocated groups I and II
- Gp I A+ B+ / Gp II B- A-
- Two signal supplies are provided from a 5/2 valve one is available only in group I the other is available only in group II

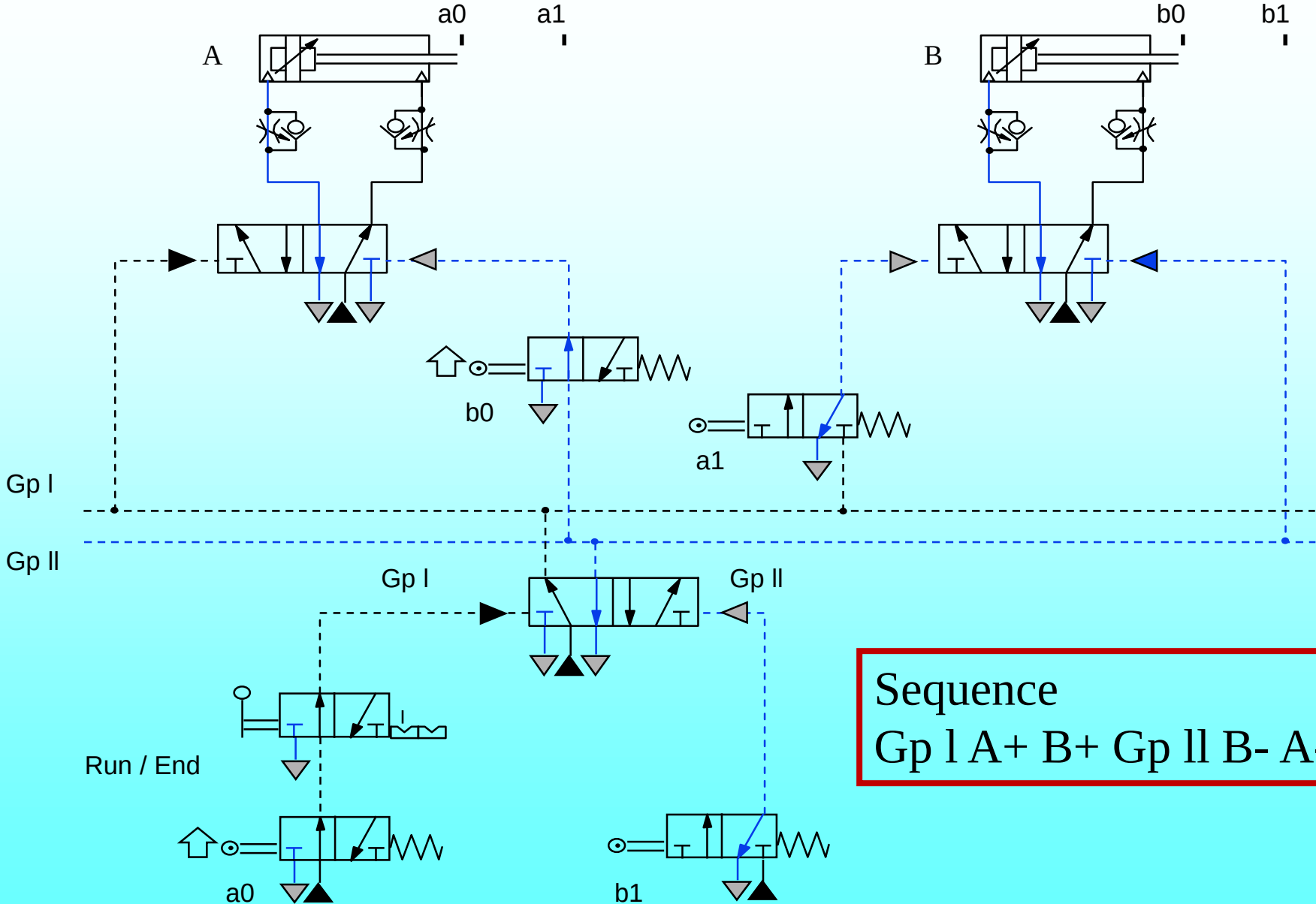
- Because only one group output is available at a time it is not possible to have opposed signals
- A standard 5/2 double pressure operated valve is the cascade valve



CASCADE (TWO GROUP)

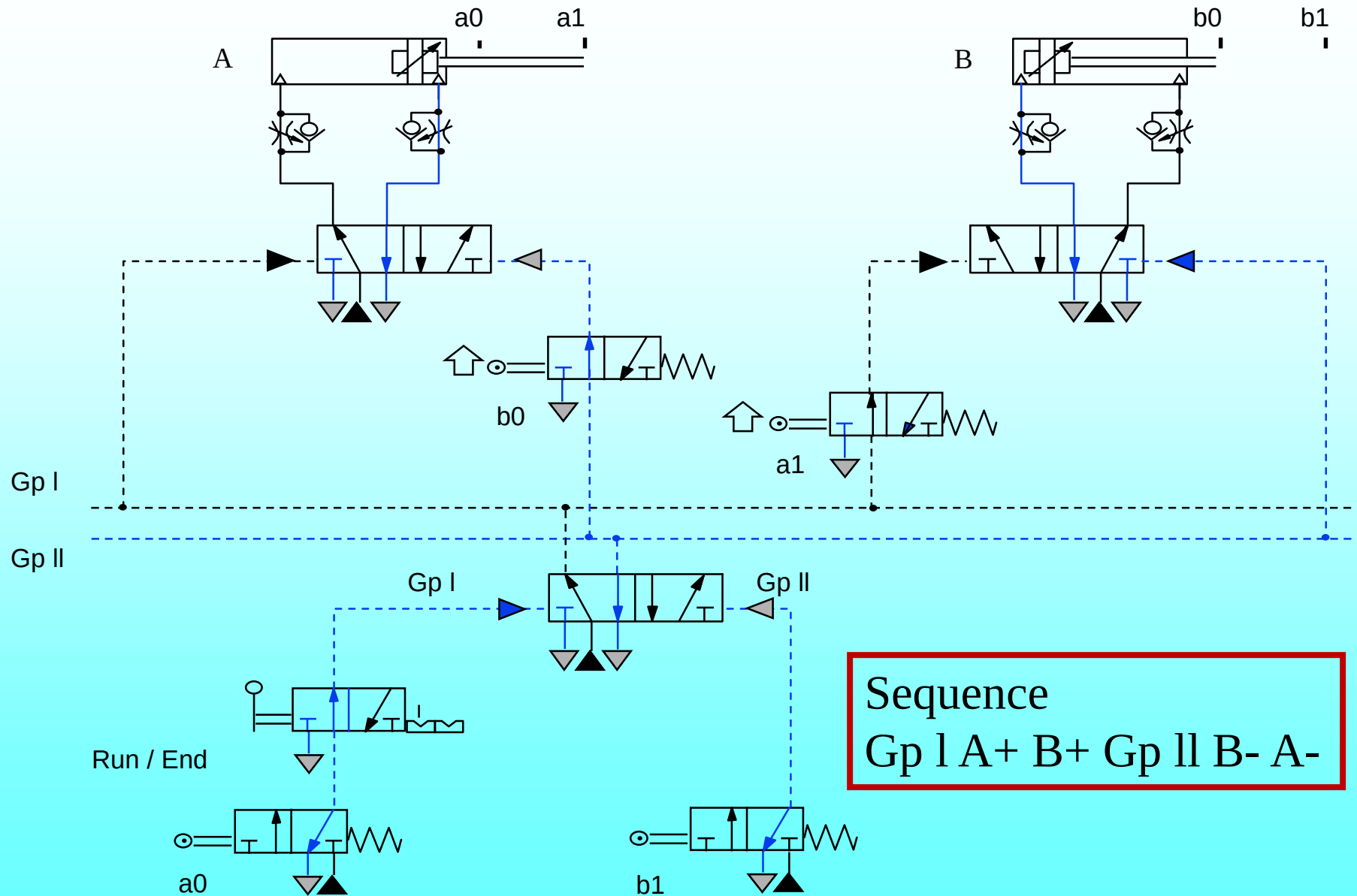


CASCADE (TWO GROUP)

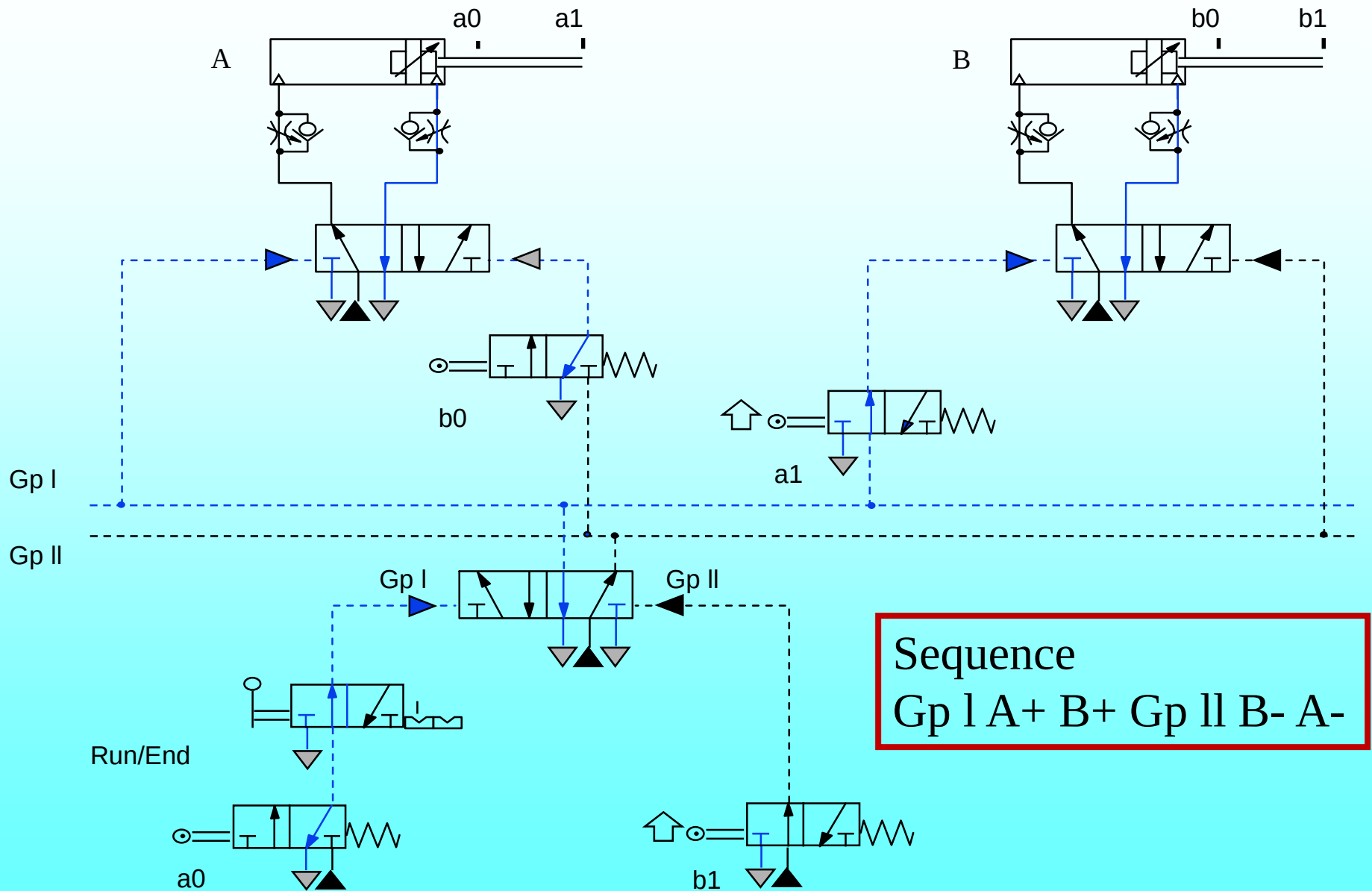


Sequence
Gp I A+ B+ Gp II B- A-

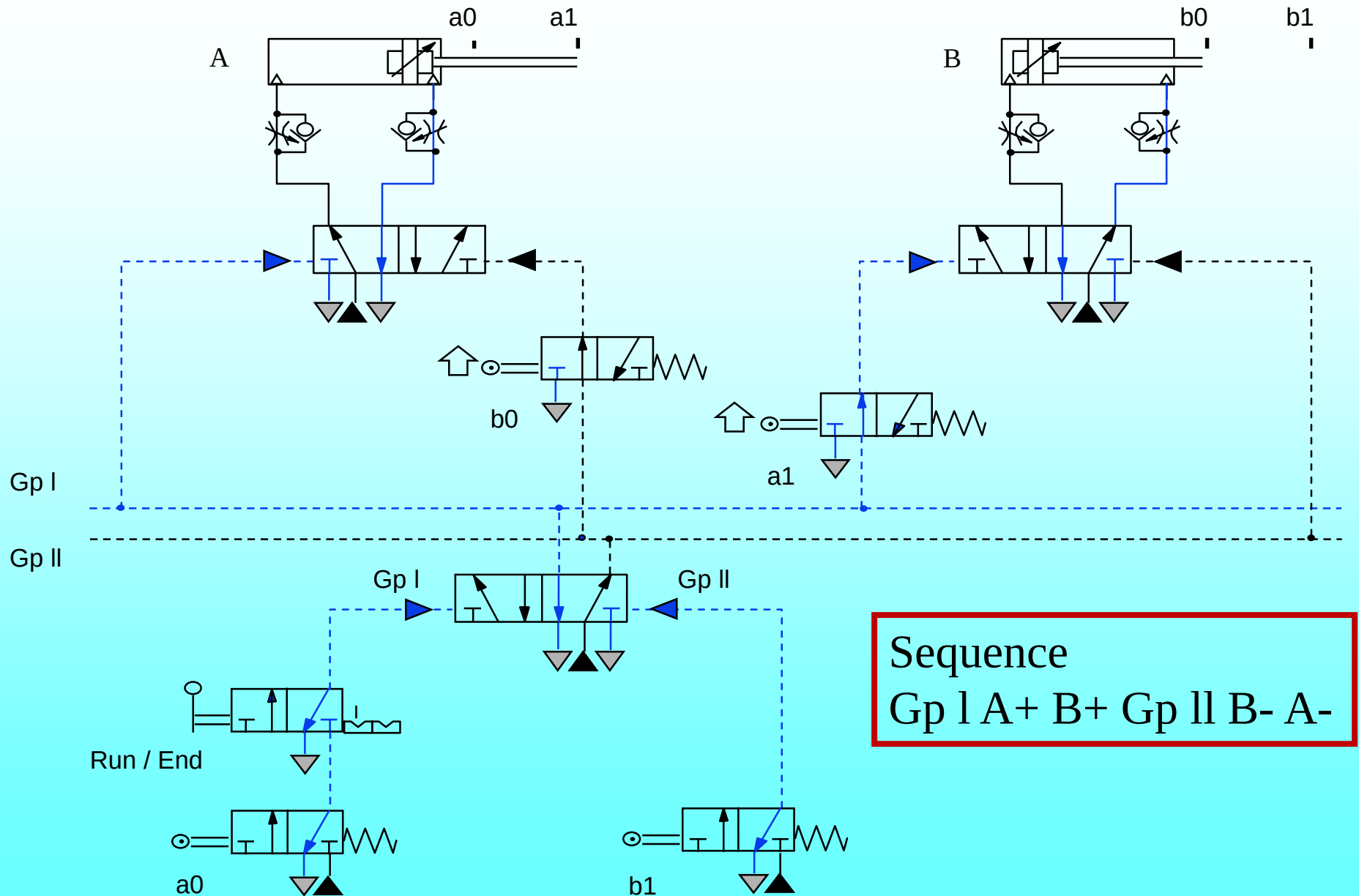
CASCADE (TWO GROUP)



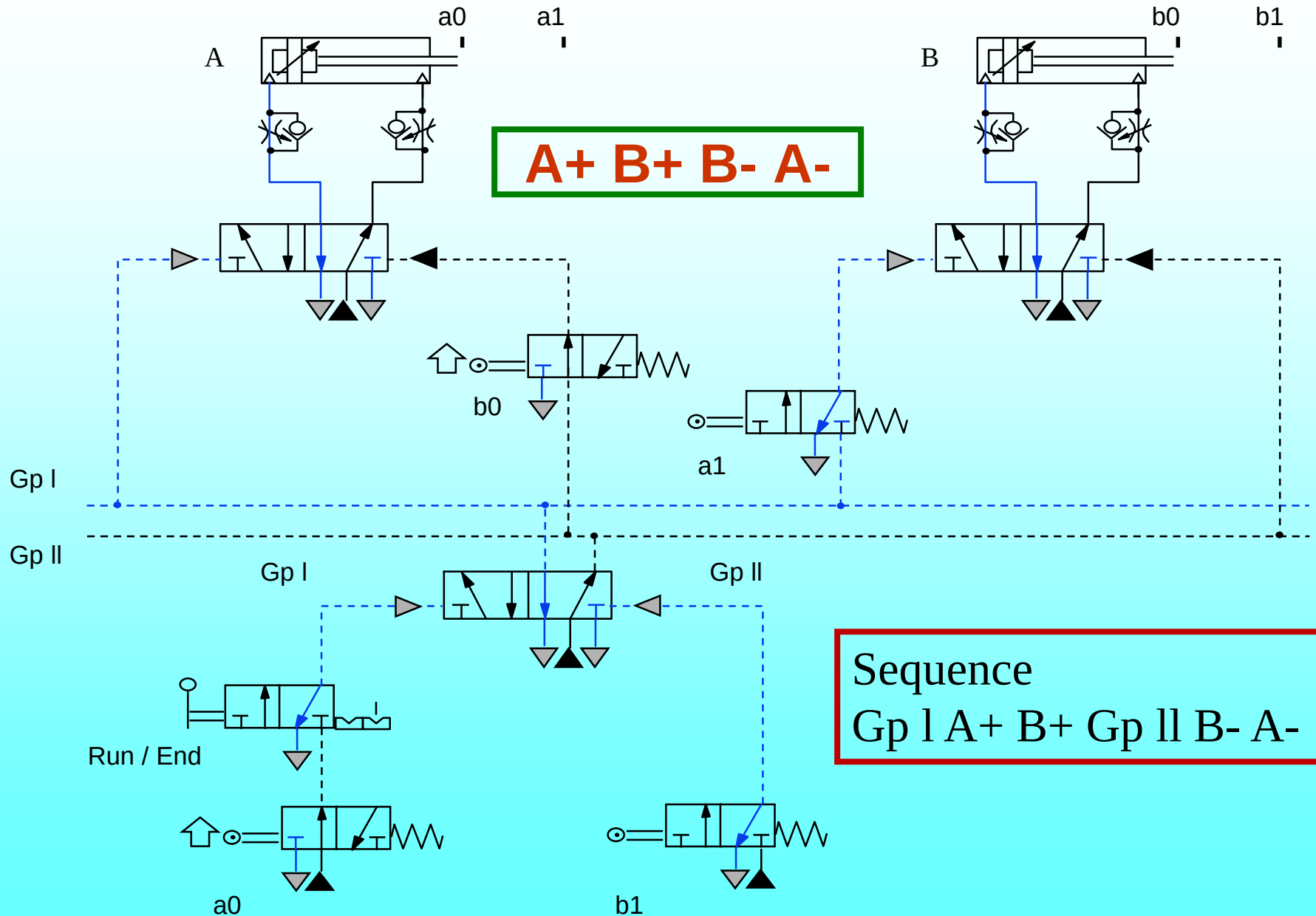
CASCADE (TWO GROUP)



CASCADE (TWO GROUP)

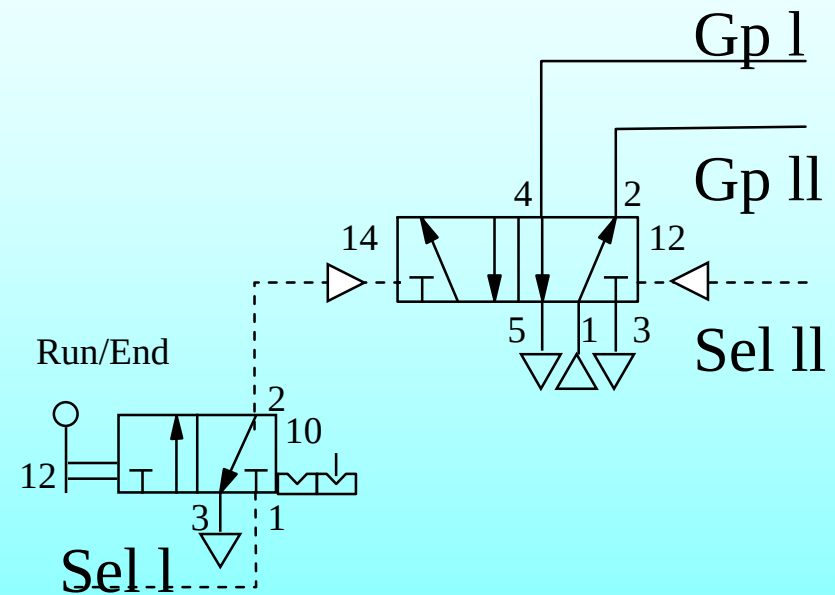


CASCADE (TWO GROUP)



CASCADE BUILDING BLOCKS

- A two group building block consists of a lever valve to run and end the sequence plus the 5/2 double pilot operated cascade valve
- For a two group system consisting of any number of cylinders this building block and the cylinder building blocks are all that is required to solve the sequence



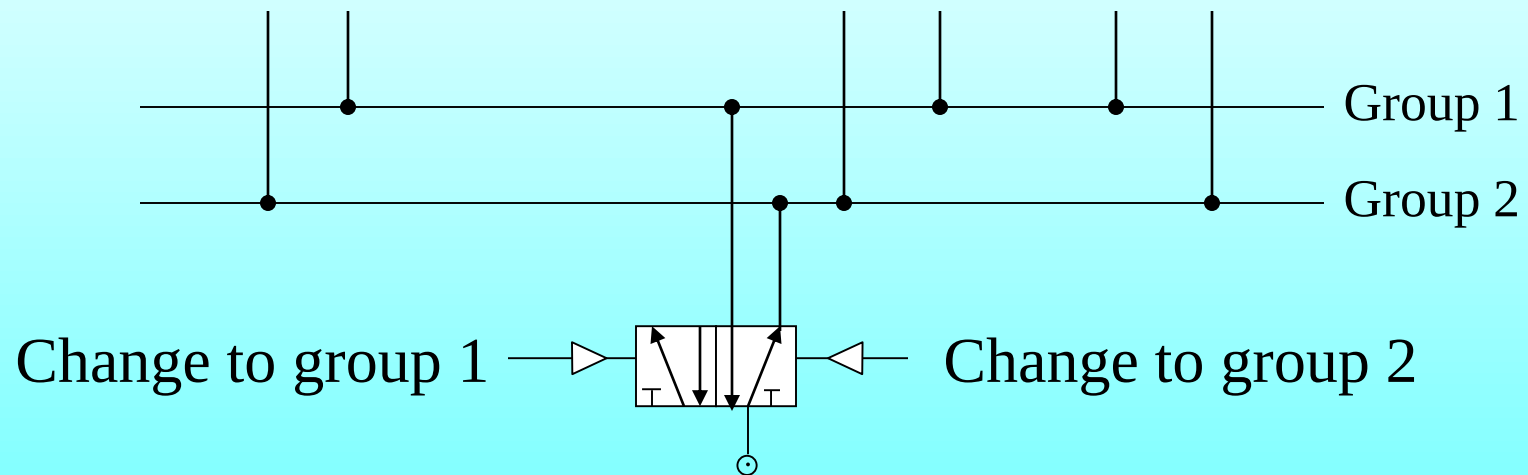
CASCADE RULES

- Establish the correct sequence
- Divide the sequence in to groups. Always start a sequence with the Run/End valve selecting group | e.g.
R/E | A+ B+ | B- C+ | C- A-
- Select the cylinder building blocks
- Select the cascade building block
- Select dual trip building blocks if required
- Interconnect the blocks as follows:
- The first function in each group is signalled directly by that group supply
- The last trip valve operated in each group is supplied with main supply air and selects the next group
- The remaining trip valves are supplied with air from their respective groups and initiate the next function
- The “run/end” valve will control the signal from the last trip valve to be operated

DESIGN OF CASCADE

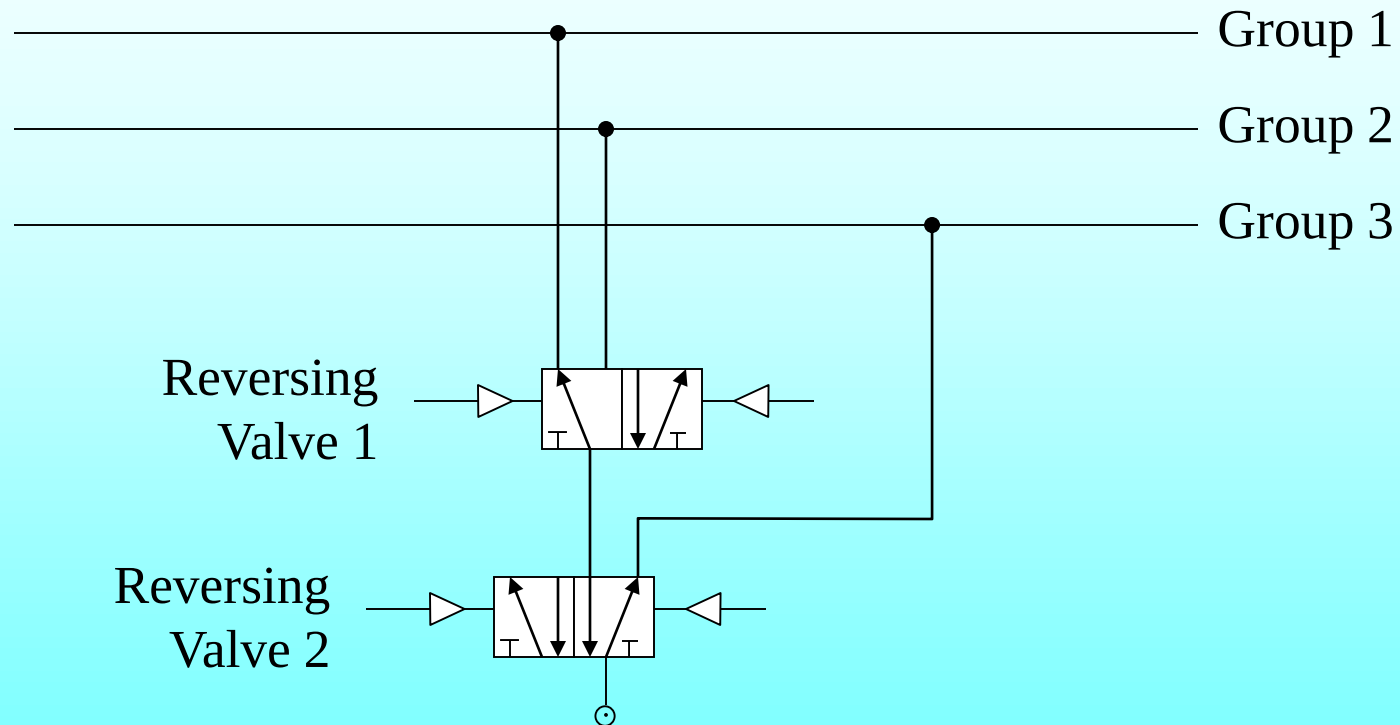
- Only one output line is pressurized at any one time

2 Groups & 1 Reversing valve



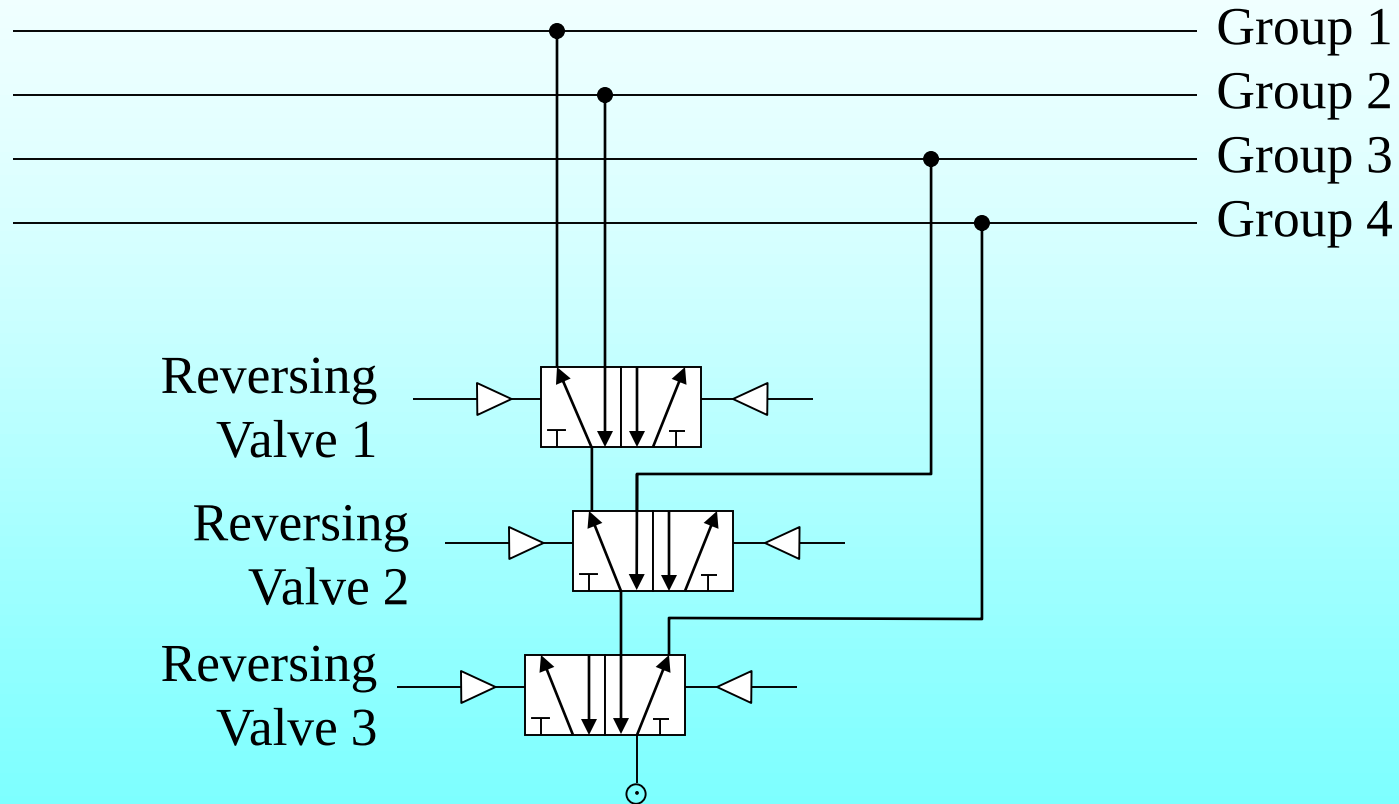
DESIGN OF CASCADE

3 Groups & 2 Reversing valves



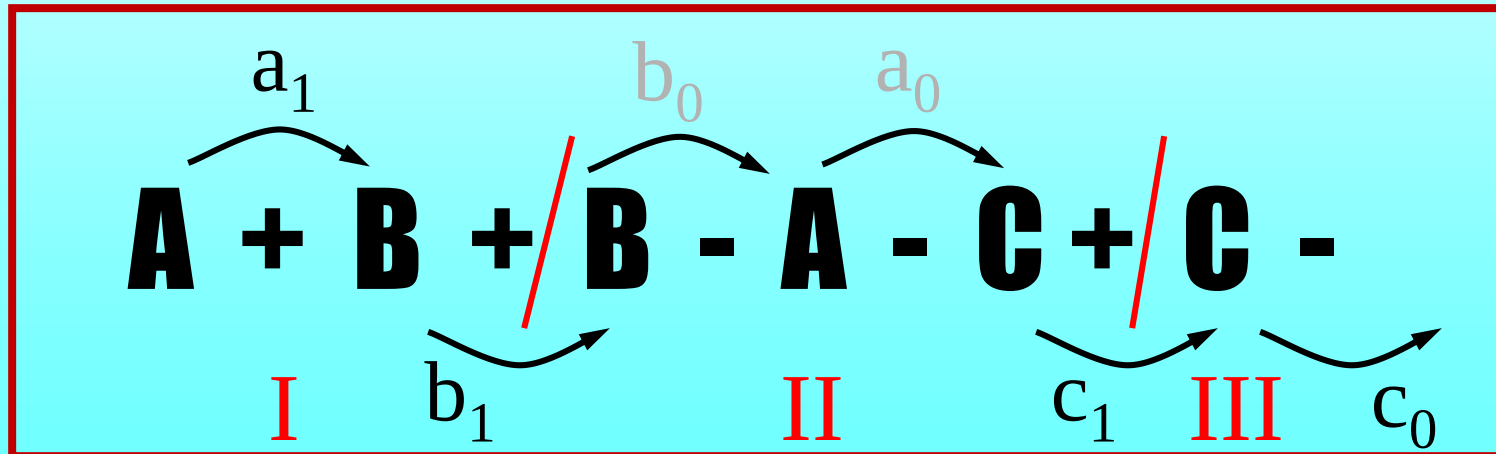
DESIGN OF CASCADE

4 Groups & 3 Reversing valves



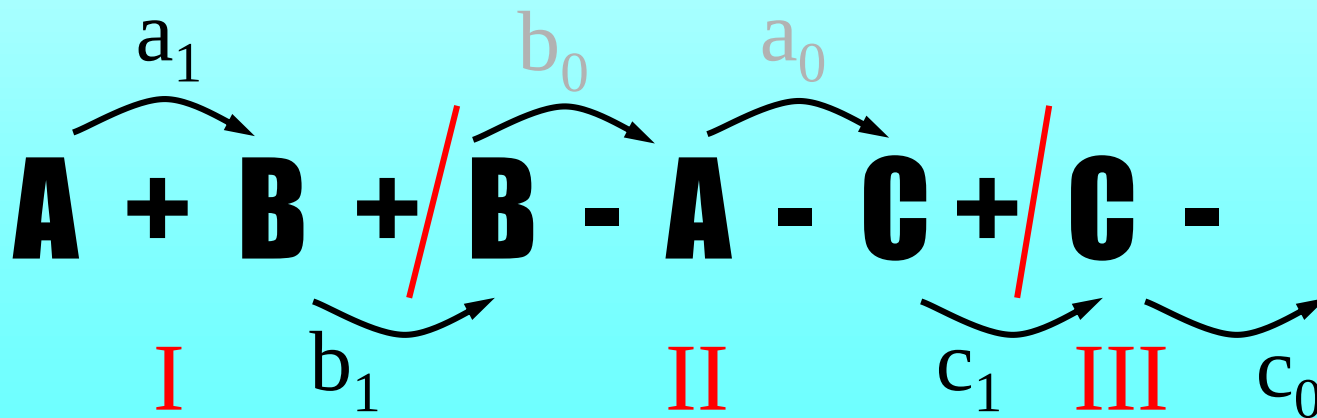
PROCEDURE FOR CONSTRUCTING A CASCADE SYSTEM

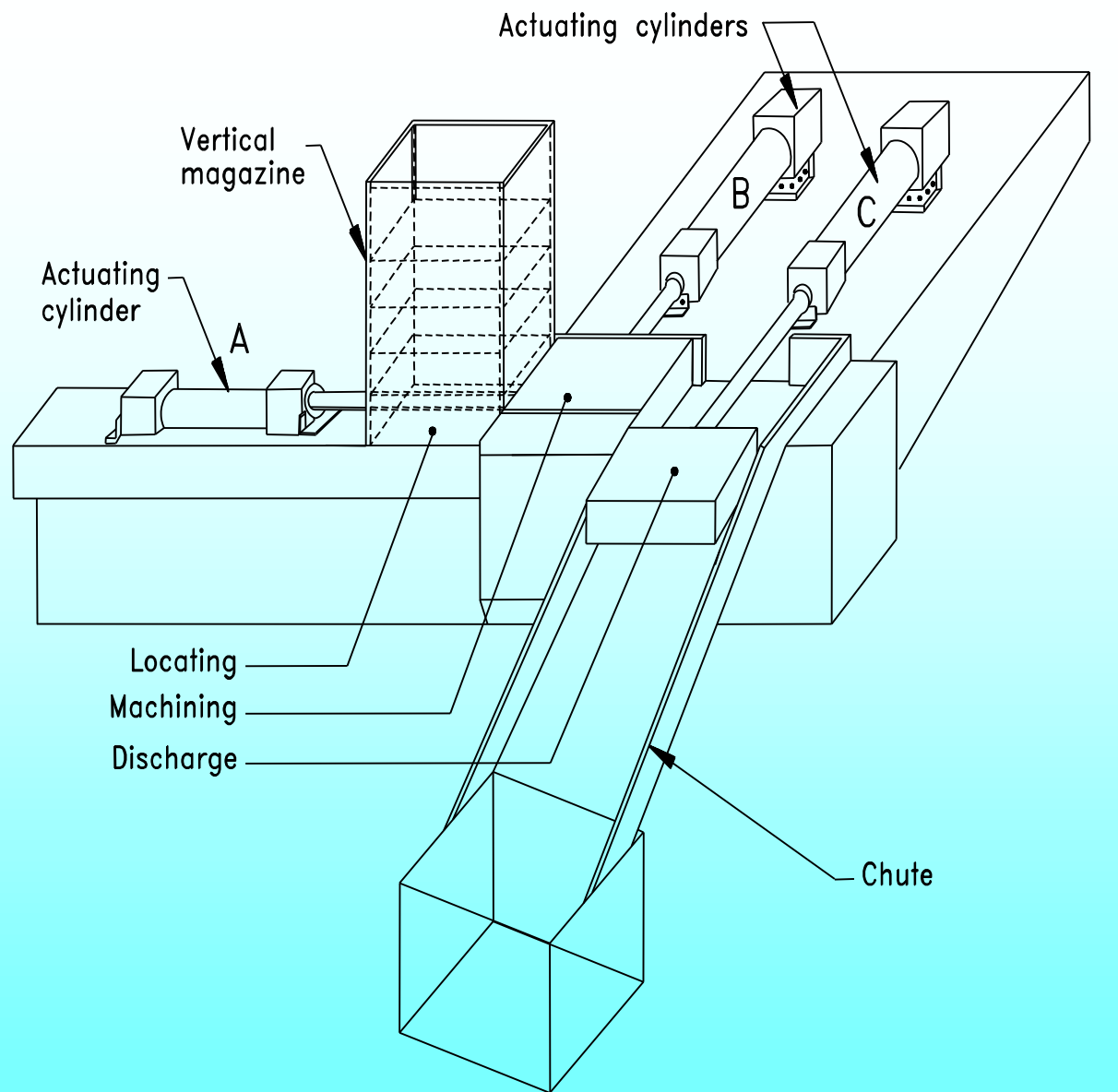
- 1) Define the sequence of motion
- 2) Divide them into minimum group
- 3) Supply lines
- 4) Reversing valves (Group Change Over valves)
- 5) Limit switches



TRANSPOSITION INTO CIRCUIT

- 1) Limit switches/valves which cause change over of groups are drawn below the supply lines
- 2) Limit switches within the same group are drawn above the supply lines
- 3) Include an “AND” valve when a cylinder extend more than one time (eg. $A + B + B - A - C + A + A - C -$)
- 4) Auxiliary conditions and additional interlocks are added only after the basic sequence design has finished





**LOCATING,
CLAMPING
and
EJECTING
the work
piece**

Configuration of Locating, clamping & ejecting the workpiece

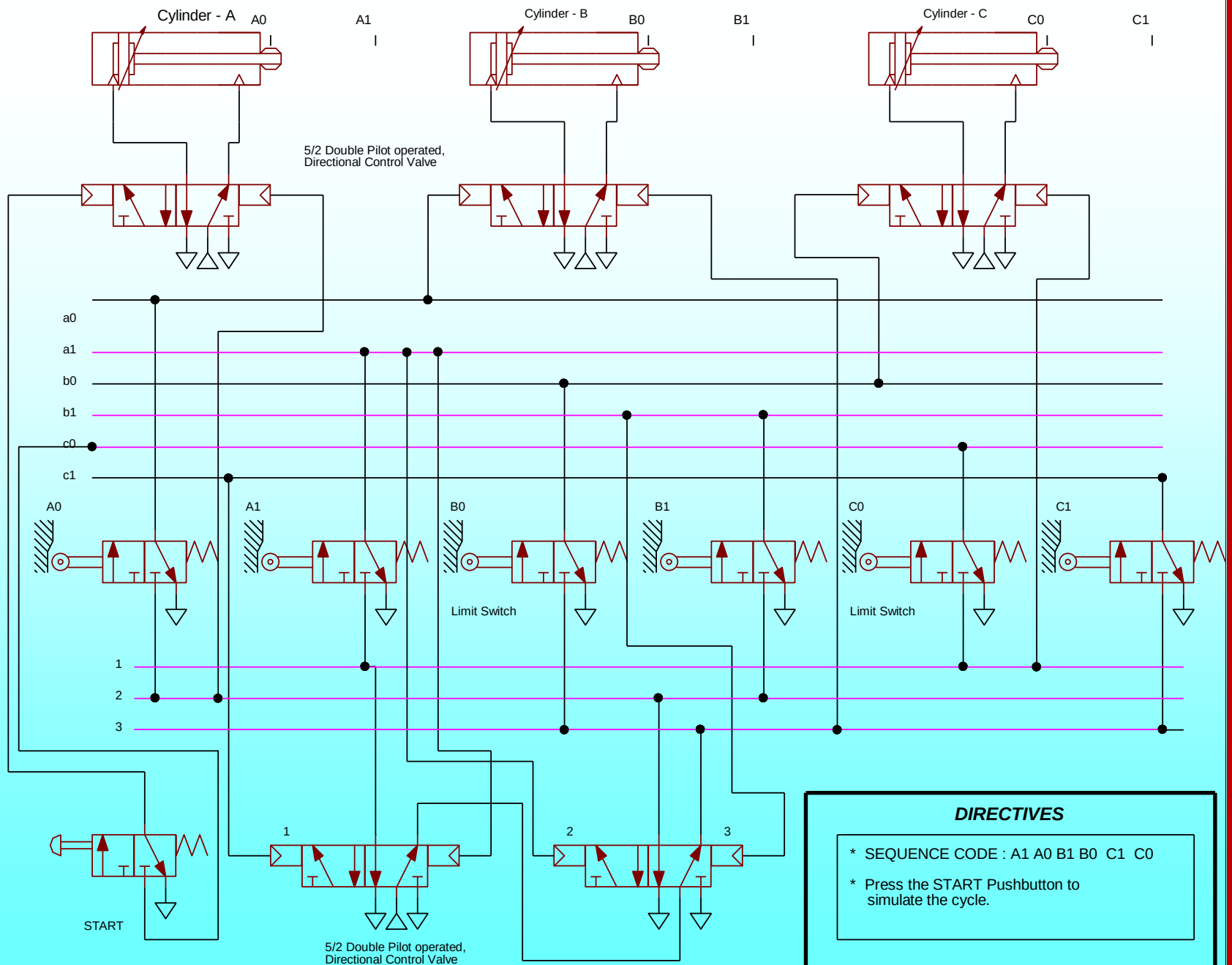
CASCADE (THREE GROUP)

Problem:

Location, clamping and ejection of a work piece are to be performed using the cylinders A,B and C. The parts are pushed by cylinder (A) and located in the processing station. Then cylinder (B) clamps the work piece for further machining. After machining, cylinder (B) unclamps and cylinder (C) ejects the work piece onto a chute. The cycle is to start when a start button is pressed. Limit switches are used to confirm cylinder positions.

POSSIBLE SEQUENCE OF OPERATIONS ARE:

A1 A0 B1 B0 C1 C0 ; A1 B1 C1 C0 B0 A0



ADVANTAGES & DISADVANTAGES OF CASCADE SYSTEM

ADVANTAGES

- Circuit can be designed and checked quickly
- Easier to trace faults
- System ensure that signals are inherently of sufficient duration to perform their require tasks

DISADVANTAGES

- May not yield the simplest solution for certain circuits
- the pressure drop becomes significant as the number of stages increase
- great care to be exercised when limit switch is depressed by external source